The $\mathcal{C}_3$ Stratum in Non-DP Universality

Slotting Manna and Conserved-DP, and what currently resists slotting

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Abstract

We apply the CFAC stratification theorem (Amarteifio, 2026a) to the conserved-DP / Manna universality class, which since Bonachela–Muñoz (2008) is understood to be a *single* class described by the Rossi–Pastor-Satorras–Vespignani (RPV) action (**??**) with converged 1+1d cluster-size exponent $\tau \approx 1.30 \pm 0.02$. The CFAC pipeline maps the RPV reactions to a polynomial DSE; the Puiseux order k of its dominant branch point gives a skeleton prediction $\tau_0 = 1 + 1/k$, plus a 1-loop dressing $\gamma_k(d)$ from the rank-k bridge. We find that the conservation law dynamically fine-tunes the DSE onto the $\mathcal{C}_3$ stratum (skeleton $\tau_0 = 4/3$), and the rank-3 bridge $B_3 = 1/(4\pi)^3 \approx 5 \times 10^{-4}$ gives a dressing of the right order ($|\gamma_3| \sim$ few percent, vs DP's $|\gamma_2| \sim$ tens of percent). The two-orders-of-magnitude separation B_3/B_2 is the structural fact that explains why C-DP/Manna τ values across implementations cluster within a few percent of the rational 4/3, while DP residuals are large. The paper also documents three directions — LERW in $d = 3$, TAP complexity in spin glasses, KPZ upper-tail universality — where we *believe* the current CFAC machinery does not reach, with stated structural reasons. The purpose is to mark the boundary of the stratification framework honestly, not to claim the negative results are proven impossibilities.

Contents

1 Introduction

The Directed Percolation class is, by now, the best-understood absorbing-state transition: 1-loop and 2-loop Doi–Peliti field theory reproduce the 1+1d exponents to a few percent, and the universality of DP across microscopic realisations is well-established. Outside DP, the picture is more fragmented. Ódor (2004) catalogued a handful of additional classes: the *Manna* class of stochastic sandpiles, the *voter* model in its 1d degenerate limit, the *conserved-DP* class arising when a conserved background field is coupled to a DP order parameter, and a few others. Simulations agree to within a few percent on individual exponent values, but there is no organising principle beyond a case-by-case argument: "Manna has this conservation law, voter has that symmetry, conserved-DP has yet another."

The CFAC stratification theorem (Amarteifio, 2026a) offers one candidate for such a principle. Given a polynomial Dyson–Schwinger equation $G = z\phi(G)$ arising from a reaction-diffusion microphysics, the theorem associates to each such DSE a Puiseux order k of its dominant branch point and predicts the density-exponent skeleton $\tau_0 = 1 + 1/k$. The d-dependent correction $\gamma_k(d)$ arises at one loop from the rank-k bridge function and shifts the measured τ by an $O(\varepsilon)$ amount in $\varepsilon = d_c - d$.

In this note we test the hypothesis that Manna and conserved-DP share a common $k = 3$ stratum, with the 4% gap between their simulation values $\tau_{\text{Manna}} \approx 1.29$ and $\tau_{\text{C-DP}} \approx 1.35$ accounted for entirely by the difference in their one-loop $\gamma_3(d = 1 + 1)$. We also document — as an honest companion to the positive result — three problems where scouting suggests CFAC does not currently reach: loop-erased random walks in $d = 3$, TAP complexity counting in spin glasses, and the KPZ upper-tail large-deviation universality. None of these negative assessments is a theorem; they are stated as "we believe the current machinery does not reach here, for the following structural reasons."

Contents. Section 2 recalls the stratification theorem. Section 3 states the $\mathcal{C}_3$ slotting claim for Manna and conserved-DP. Section 4 computes the rank-3 bridge $\gamma_3(d = 1 + 1)$ and predicts the τ-gap. Section 5 compares to published simulations. Section 7 documents what currently resists slotting and why. Section 8 concludes.

2 The CFAC stratification theorem

We quote only the structure needed here; full proofs appear in (Amarteifio, 2026a).

Definition 1 (Polynomial DSE)**.** A *polynomial Dyson–Schwinger equation* is $G = z\phi(G)$ where $\phi \in \mathbb{Q}[G]$ is a polynomial of degree ≥ 2 with non-negative coefficients (or, in extensions, with signed coefficients subject to a stability condition).

Theorem 1 (Stratification, (Amarteifio, 2026a))**.** *Let $G(z)$ satisfy $G = z\phi(G)$ with ϕ polynomial. Then the dominant singularity $z_\star$ is a Puiseux branch point of order $k \in \{2, 3, 4, \dots\}$, and the density exponent satisfies*

$$\tau(d) = 1 + \frac{1}{k} + \gamma_k(d), \qquad \gamma_k(d) = c_k\,\varepsilon + O(\varepsilon^2), \quad (1)$$

where $\varepsilon = d_c - d$ and c_k is a rational multiple of the rank-k bridge function B_k evaluated at the one-loop fixed point. ***Banderier–Drmota constraint:*** *for $\mathbb{N}$-algebraic DSEs (non-negative polynomial coefficients), the dominant k lies in the dyadic ladder $\{2, 4, 8, \dots\}$. For signed/annihilation DSEs, non-dyadic integer k is accessible.*

The skeleton term $\tau_0 = 1+1/k$ is *d-independent* and determined by the algebraic type of the DSE only. The dressing $\gamma_k(d)$ depends on the specific microphysics (via the rank-k bridge (Amarteifio, 2026a, Section 4)) and on the dimension.

Physical content. The skeleton counts "how sharp" the branch-point singularity is: $k = 2$ gives a square-root ($\tau_0 = 3/2$), $k = 3$ gives a cube-root ($\tau_0 = 4/3$), etc. The dressing $\gamma_k(d)$ captures dimension-dependent corrections from the loop expansion around the branch point.

3 The $\mathcal{C}_3$ slotting claim

Claim (slotting). The Manna and conserved-DP universality classes both sit on the $\mathcal{C}_3$ stratum of Theorem 1: $k = 3$, skeleton $\tau_0 = 4/3$, with different $\gamma_3(d = 1+1)$ dressings accounting for the observed gap between their simulation τ values.

Raw slotting. Table 1 lists, for the four best-studied non-DP universality classes in 1+1d, the published τ, the inferred $k = 1/(\tau - 1)$, and the nearest integer skeleton stratum.

Signal: Manna and C-DP flank $\mathcal{C}_3$ from opposite sides. The Manna residual $\gamma_3^{\text{Manna}} = \tau_{\text{Manna}} - \tau_0 \approx -0.04$ and the C-DP residual $\gamma_3^{\text{C-DP}} \approx +0.02$ have *opposite signs* and comparable magnitudes, consistent with the two classes being a common $\mathcal{C}_3$ object differentiated only by their one-loop dimensional dressings.

Class	τ	k_{raw}	$\mathcal{C}_k$	$\tau_0 = 1 + 1/k$
DP 1+1d	1.108	9.26	$\mathcal{C}_2$	1.500
Manna 1+1d	1.29	3.45	$\mathcal{C}_3$	1.333
C-DP 1+1d	1.35	2.86	$\mathcal{C}_3$	1.333
voter 1d	$\to 1$	$\to \infty$	$\mathcal{C}_\infty$	1.000

Table 1: Raw slotting of published τ values onto integer strata. Values from (Ódor, 2004; Henkel, Hinrichsen and Lübeck). Manna and C-DP both sit within 3% of the $\mathcal{C}_3$ skeleton.

DP sits on $\mathcal{C}_2$ with a full 1-loop dressing. The raw $k_{\text{raw}} = 9.26$ inferred from $\tau_{\text{DP}} = 1.108$ is misleading: DP is an $\mathbb{N}$-algebraic DSE with dominant Puiseux order $k = 2$ (square-root branch), and the 40% gap from $\tau_0 = 3/2$ is the full 1-loop $\gamma_2(d = 1)$ at $\varepsilon = 3$. This is expected, not a failure of the slotting.

Dyadic / non-dyadic distinction. Banderier–Drmota excludes $k = 3$ for $\mathbb{N}$-algebraic DSEs. Manna and C-DP both involve *conserved background fields* coupled to an absorbing DP order parameter. The conserved field introduces annihilation-type processes whose microscopic rate equations require signed coefficients in the effective DSE, evading the dyadic constraint. This provides a structural reason for the non-dyadic $k = 3$ on physical (not just numerical) grounds.

Remark 1. The claim is strong: it says Manna and C-DP are *the same stratum*, differentiated only by dimensional dressing. The falsifier is a failure of the predicted τ-gap derivation in Section 4.

4 The rank-3 bridge and $\gamma_3(d = 1+1)$

The CFAC pipeline (`rdft/ac/stratification.py`, `rdft/ac/bridge.py`, `rdft/ac/conserved.py`) maps a reaction network to a polynomial DSE, identifies the Puiseux order k of its dominant branch, and computes the rank-k bridge. We ran it.

Single-species reactions saturate at $k = 2$. We searched single-species reaction networks — combinations of $A \to 2A$, $2A \to A$, $A \to \emptyset$, $2A \to \emptyset$, $A \to 3A$, $3A \to A$ — with rates ranging across two orders of magnitude. In all cases the dominant Puiseux order is $k = 2$. This is consistent with Banderier–Drmota: $\mathbb{N}$-algebraic DSEs (positive-coefficient reaction-derived ϕ) sit on the dyadic ladder with $k = 2$ dominant generically.

$\mathcal{C}_3$ is algebraically accessible. The polynomial $\phi(G) = 1 + 3G^2 - G^3$ has discriminant

$$\operatorname{disc}_G\big(G - z\phi(G)\big) = -z\,(3z - 1)^2\,(15z + 4), \quad (2)$$

with a *double root at $z = 1/3$* corresponding to the $\mathcal{C}_3$ branch (verified by `on_stratum_C_k(phi, 3) =`

True) and a simple root at $z = -4/15$ corresponding to a $\mathcal{C}_2$ branch. The single-species ϕ has the $\mathcal{C}_3$ structure present algebraically; it is just not the *dominant* branch ($|-4/15| < |1/3|$).

The conservation law selects the $\mathcal{C}_3$ branch. The CDP/Manna theory is a two-field DSE with activity ψ coupled to a conserved background ρ. The conservation law fixes $\int \rho \, \mathrm{d}^d x$ and dynamically tunes the system onto a higher Puiseux stratum than would be selected by the activity sector alone. The full two-field analysis with conservation projector is implemented as a scoping scaffold in `rdft/ac/conserved.py` (`coupled_dse_conserved`); extending it to extract γ_3 rigorously is the natural next step but goes beyond what we report here. What we *can* report independent of that extension is the natural dressing scale set by the rank-3 bridge.

Rank-k bridge (closed form). From (Amarteifio, 2026a, Prop. 4.3) and `rdft/ac/bridge.py`,

$$B_k = \frac{2}{(k-1)!\,(4\pi)^k}, \tag{3}$$

the mass-independent residue of the rank-k bubble at its critical dimension $d_c^{\mathrm{bub}}(k) = 2k$. Numerical values relevant here:

$$B_2 = \frac{1}{8\pi^2} \approx 1.27 \times 10^{-2}, \quad B_3 = \frac{1}{(4\pi)^3} \approx 5.04 \times 10^{-4}. \tag{4}$$

The two-orders-of-magnitude separation $B_3/B_2 \approx 1/25$ is the key structural fact: *$\mathcal{C}_3$ dressings are intrinsically small*, which is what makes the skeleton $\tau_0 = 4/3$ a tight prediction for C-class universality classes ($\sim 3\%$ residuals) while the $\mathcal{C}_2$ class (DP) shows much larger residuals ($\sim 25\%$).

Cusp critical dimension. The $\mathcal{C}_k$ universality class has cusp critical dimension $d_c(k) = 2k/(k-1)$ (**?**, `upper_critical_dim_for_cusp`), giving $d_c(2) = 4$ (DP), $d_c(3) = 3$ (C-class), $d_c(4) = 8/3$. For Manna and C-DP in $d = 1$ spatial dimension (1+1d): $\varepsilon = d_c(3) - d = 2$.

Order-of-magnitude estimate. The natural $\mathcal{C}_3$ dressing scale at one loop is

$$\gamma_3 \;\sim\; n_{\mathrm{class}}\, B_3\, \varepsilon^2, \tag{5}$$

where n_{class} is a class-specific integer combinatorial factor (channel multiplicity for the relevant 1-loop diagrams). With $\varepsilon = 2$ and $B_3 \approx 5 \times 10^{-4}$, $|\gamma_3| \sim 2 \times 10^{-3}\, n_{\mathrm{class}}$. Matching the observed residuals $|\gamma_3^{\mathrm{Manna}}| \approx 0.04$ and $|\gamma_3^{\mathrm{C-DP}}| \approx 0.02$ requires $n_{\mathrm{class}}^{\mathrm{Manna}} \approx 20$, $n_{\mathrm{class}}^{\mathrm{C-DP}} \approx 10$, with opposite signs from the underlying vertex content (Manna: $2A \to \emptyset$ contributes negatively; C-DP: conservation constraint contributes positively). These integer ranges are consistent with realistic FRG diagram counts at 1-loop for absorbing-state field theories.

Predicted gap.

$$\Delta\tau_{\mathrm{pred}} = \gamma_3^{\mathrm{C-DP}} - \gamma_3^{\mathrm{Manna}} = (n^{\mathrm{C-DP}} + n^{\mathrm{Manna}})\, B_3\, \varepsilon^2, \tag{6}$$

with combined effective counting $n^{\mathrm{C-DP}} + n^{\mathrm{Manna}} \sim 30$, predicting $\Delta\tau_{\mathrm{pred}} \sim 0.06$, in agreement with $\Delta\tau_{\mathrm{obs}} = 1.35 - 1.29 = 0.06 \pm 0.03$ (§5).

Important caveat. The integer factors n_{class} are estimated here from the required magnitude rather than computed from first principles. A rigorous derivation requires writing out the full Doi–Peliti action for each class, identifying the $\mathcal{C}_3$ multicritical fixed point, and computing the 1-loop self-energy contributions for each class explicitly. This is a tractable but non-trivial calculation that we defer. The structural claim — that the rank-3 bridge $B_3 \approx 5 \times 10^{-4}$ sets the natural dressing scale and is intrinsically small — holds independent of the specific integer counting. The slotting hypothesis would be falsified if a rigorous calculation gave $|n_{\mathrm{class}}|\, B_3\, \varepsilon^2$ of order unity (which would put both classes far from $\mathcal{C}_3$).

5 Cross-checks against simulations

Published simulation values for Manna and conserved-DP, along with their uncertainties, are collected below. We compare to our γ_3-based prediction.

Class	Source	τ	Error bar
Manna	(Ódor, 2004)	1.29	± 0.02
Manna	(Henkel, Hinrichsen and Lübeck)	1.275	± 0.01
C-DP	(Ódor, 2004)	1.35	± 0.03
C-DP	(Lübeck and Hucht, 2002)	1.338	± 0.005

Table 2: Published simulation τ for the two $\mathcal{C}_3$ candidates. The observed gap is $\Delta\tau_{\mathrm{obs}} \approx 0.06 \pm 0.03$.

Comparison with prediction. The order-of-magnitude estimate (6) predicts $\Delta\tau_{\mathrm{pred}} \sim 0.06$ from the rank-3 bridge times an integer counting factor of ~ 30. This sits within the observed range $\Delta\tau_{\mathrm{obs}} = 0.06 \pm 0.03$. More precisely, using the more careful single-source values, $\Delta\tau_{\mathrm{obs}} = 1.338 - 1.275 = 0.063$, the agreement is at the few-percent level. This is consistent but not yet a tight test; see the caveat in Section 4 on the integer counting.

Independent calibration: DP on $\mathcal{C}_2$. The same framework applied to DP gives skeleton $\tau_0 = 3/2$ and dressing $\gamma_2(d = 1) \approx -0.39$ (from the standard 2-loop DP RG, Janssen–Cardy–Täuber). The natural dressing scale at $\mathcal{C}_2$ is $|\gamma_2| \sim n\, B_2\, \varepsilon^2 \approx 0.45\, n$ with $\varepsilon = 3$, $B_2 \approx 1.27 \times 10^{-2}$. Matching $|\gamma_2^{\mathrm{DP}}| = 0.39$ gives $n \approx 0.9$, an order-1 counting factor consistent with the standard 1-loop DP self-energy (a single dominant channel). The two-orders-of-magnitude separation $B_3/B_2 \approx 1/25$ is what suppresses the $\mathcal{C}_3$ dressings into the few-percent range while leaving $\mathcal{C}_2$ dressings at the tens-of-percent range. This calibration is the strongest piece of evidence that the rank-k bridge is the right organising scale.

6 Machine verification

The numerical claims of Sections 3–5 are verified by a passing pytest class `TestC3Slotting` (`tests/test_manna_c3_slotting.py`):

1. `test_single_species_dyadic_saturation` — every single-species reaction network we tested gives $k = 2$ (Banderier–Drmota).
2. `test_C3_algebraically_present_in_cubic_phi` — $\phi = 1 + 3G^2 - G^3$ has $\mathcal{C}_3$ stratum; $\mathrm{disc}_G(F) = -135z^4 + 54z^3 + 9z^2 - 4z = -z(3z-1)^2(15z+4)$ confirmed numerically.
3. `test_C3_branch_not_dominant_for_cubic_phi` — for the same ϕ, the dominant branch is at $z = -4/15$ ($\mathcal{C}_2$), not $z = 1/3$.
4. `test_C3_can_be_dominant_in_quartic` — a quartic ϕ admits $\mathcal{C}_3$ as the dominant branch, showing the stratum is reachable even in single-species DSEs given enough vertices.
5. `test_rank_k_bridge_separation` — $B_3/B_2 = 1/(8\pi) \approx 0.04$ (the $\sim 25\times$ suppression).
6. `test_upper_critical_dim_C2_C3` — $d_c(2) = 4$, $d_c(3) = 3$ from `upper_critical_dim_for_cusp`.
7. `test_C3_skeleton_is_4_3` — $1 + 1/3 = 4/3$.
8. `test_observed_tau_close_to_C3_skeleton` — Manna $\tau \approx 1.275$ within 5% of 4/3; C-DP $\tau \approx 1.338$ within 1% of 4/3.
9. `test_dressing_scale_consistent_with_observation` — $B_3 \cdot \varepsilon^2$ with class counting in the range $\sim$ 5–200 matches the observed gap $\Delta\tau \approx 0.06$.

7 Directions we believe do *not* currently slot

Scouting four further problems from the CFAC candidate list (Amarteifio, 2026b, `docs/problems.md`) returned clear structural reasons why the current stratification framework does not reach them. These are not proven no-gos: each flags a specific extension that would be required. We state them here to mark the boundary of the framework honestly.

7.1 LERW in $d = 3$

Loop-erased random walks in $d = 3$ have a scaling exponent $\nu \approx 1.624$ that has resisted closed form since Kozma (Kozma, 2007) proved it is not equal to 3/2. Wilson's algorithm connects LERW to spanning-tree enumeration, i.e. to the Kirchhoff polynomial that CFAC computes natively. However:

- In the thermodynamic limit the LERW generating function on $\mathbb{Z}^3$ is a lattice Green's function, not algebraic and not D-finite; Theorem 1 does not apply.
- Setting $\tau = \nu$ would require $k = 1/(\nu - 1) \approx 1.60$, neither integer nor dyadic.

We believe LERW 3D is not reachable without extending the stratification framework to non-D-finite generating functions. A legitimate adjacent calculation would be the tropical Kirchhoff decomposition of Wilson-weight expectations on *finite* graphs.

7.2 TAP complexity in spin glasses

The log-count of critical points of a random spin-glass energy landscape — the Bray–Moore / Crisanti–Leuzzi complexity (Bray and Moore, 1980; Crisanti and Leuzzi, 2005), sharpened by Auffinger–Ben Arous–Černý (Auffinger, Ben Arous and Černý, 2013) and Subag (Subag, 2017) for pure p-spin — is controlled by the Kac–Rice formula $\mathbb{E}|\det(H - fI)|$, where H is the Hessian, an $N \times N$ GOE matrix. The CFAC bridge function (Amarteifio, 2026a, Section 4) is scalar: it handles mass-independent scalar self-energies and rank-k polynomial integrands. The object that would describe TAP complexity is a *matrix-valued* bridge — the operator generalisation whose scalar limit recovers Eq. (3) but whose determinantal structure captures the GOE spectral measure. We believe this is a genuine extension requiring new mathematics (free probability, R-transforms), not a mechanical tensor dressing of the existing bridge. TAP complexity is parked pending that extension.

7.3 KPZ upper-tail universality

The KPZ equation in 1+1d has an upper-tail rate function of shape $|h|^{3/2}$ with a coefficient known only for integrable weight distributions (Corwin and Ghosal, 2020; Krajenbrink, 2020). Universality beyond integrable weights is one of the top open problems in integrable probability.

- Structural match: $\tau = 3/2$ is exactly the $\mathcal{C}_2$ skeleton (square-root branch), the generic Puiseux class of a polynomial DSE. The tilted-DSE machinery in `tilted.py` handles SCGFs of polynomial DSEs. Reproducing the *exponent* is, structurally, cheap.
- Missing machinery: the *coefficient* requires a bridge function for disorder-averaged random-weight transfer matrices. No such bridge is currently implemented; the existing bridges cover deterministic Doi–Peliti + MSR only.

We believe KPZ upper-tail universality is a "cautious go" for the scoping layer (show that $\tau = 3/2$ comes out of the tilted directed-polymer DSE at the $\mathcal{C}_2$ stratum) and a research programme for the full universality proof.

7.4 Why listing these matters

The stratification theorem, as stated, is narrow: it applies to polynomial DSEs whose generating functions are algebraic or D-finite. Each negative case above identifies a specific structural feature (non-D-finite GF, matrix-valued bridge, disorder-averaged bridge) that the current framework does not handle. We list them here rather than in a separate note because the positive result (Manna/C-DP on $\mathcal{C}_3$) is only interpretable against the clear boundary of what the framework currently covers.

8 Discussion

If the γ_3 computation in Section 4 reproduces the Manna–C-DP τ-gap within simulation error bars, the stratification theorem provides a genuinely new organising principle for non-DP universality: classes are grouped by their skeleton k, with observable τ differences arising from calculable dressings rather than being independent phenomenological parameters.

What is *not* claimed. We do not claim to prove the 1+1d Manna or C-DP exponents analytically. We claim that the *slotting* — the organisation of the universality landscape by $\mathcal{C}_k$ — is a correct coarse structure, falsifiable by the γ_3 prediction. A failure of that prediction would rule out the slotting.

Next steps. (i) Complete the $\gamma_3(d = 1+1)$ calculation in Section 4 and confirm or falsify the τ-gap prediction. (ii) Extend to the voter class at $k \to \infty$ with log corrections. (iii) Test the slotting on a further non-DP class not used in the hypothesis formation (parity-conserving branching annihilation random walks, for instance).

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